

EXECUTIVE SUMMARY

E-COMMERCE SALES PERFORMANCE ANALYSIS AND INTERACTIVE DASHBOARD DEVELOPMENT

INTRODUCTION

In today's highly competitive e-commerce environment, organizations rely on data analytics to gain actionable insights into customer behaviour, operational efficiency, and overall business performance. The ability to transform raw transactional data into meaningful information enables management to make informed strategic decisions, improve customer satisfaction, and maximize profitability.

This project focused on analysing an online retail sales dataset and developing a comprehensive interactive dashboard using Microsoft Excel. The primary objective was to evaluate sales performance, understand customer purchasing patterns, measure the effectiveness of promotional campaigns, assess marketing channels, and monitor order fulfilment activities. By combining data cleaning techniques, analytical calculations, PivotTables, PivotCharts, and dashboard design principles, the project produced an intuitive reporting solution capable of supporting executive decision-making.

PROJECT OBJECTIVES

The project was undertaken to achieve the following objectives:

- Analyse overall sales performance using key business metrics.
- Monitor customer purchasing behaviour and spending patterns.
- Measure the effectiveness of promotional coupon campaigns.
- Evaluate the contribution of different referral sources to overall revenue.
- Assess order fulfilment performance using order status information.
- Identify high-value customers for targeted business strategies.
- Develop an interactive dashboard that allows users to filter and explore business performance dynamically.

The resulting dashboard serves as a centralized reporting platform that enables management to quickly assess business performance and identify areas requiring attention.

METHODOLOGY

The project followed a structured business intelligence workflow consisting of data preparation, analysis, visualization, and reporting.

The first stage involved reviewing the dataset to ensure completeness and consistency. Missing coupon values were standardized by replacing blank entries with "NIL", allowing promotional and non-promotional orders to be clearly distinguished during analysis. The dataset was then converted into an Excel Table to improve scalability, automate calculations, and simplify the creation of PivotTables and charts.

Following data preparation, several key performance indicators (KPIs) were developed to measure business performance. These included Total Revenue, Total Orders, Total Customers, Average Order Value, Average Cart Size, Order Completion Rate, Coupon Usage Rate, and Average Coupon Order Value. These metrics provided quantitative measures for evaluating both financial performance and customer engagement.

To improve usability, the dashboard was divided into two functional sections. The first page presents an executive overview of the business through high-level KPIs and operational charts, while the second page focuses on customer behaviour, promotional effectiveness, and marketing performance. Interactive slicers were incorporated to enable users to analyse data by product, payment method, referral source, coupon code, order status, and date.

This structured analytical approach ensured that every visualization addressed a specific business question and contributed meaningful insights for decision-making.

KEY FINDINGS

The analysis revealed several important observations regarding the company's sales performance and customer behaviour.

The business generated a total revenue exceeding one million US Dollars from 1,200 recorded transactions involving over 1,100 unique customers. This indicates a healthy customer acquisition rate while also suggesting an opportunity to improve customer retention, as the number of unique customers was very close to the total number of orders.

The calculated average order value demonstrated that customers generally placed moderately valued purchases, while the average cart size indicated that multiple products were commonly

purchased within a single transaction. These findings suggest opportunities for additional cross-selling and bundled product promotions to further increase revenue per order.

Marketing analysis revealed that promotional coupons played a significant role in influencing purchasing behaviour, with a substantial proportion of completed orders making use of coupon codes. This suggests that promotional campaigns remain an effective sales driver and should continue to form part of the organization's marketing strategy. Furthermore, customers who used promotional coupons maintained healthy average spending levels, indicating that discounts did not significantly reduce transaction values.

Referral source analysis identified Instagram as the highest-performing marketing channel in terms of revenue generation. This finding highlights the importance of social media marketing and suggests that continued investment in high-performing digital channels may produce additional revenue growth. Conversely, referral marketing generated comparatively lower revenue, indicating potential opportunities to redesign referral programmes or introduce stronger customer incentives.

Customer analysis also revealed a small group of high-value customers who contributed disproportionately to total revenue. These customers represent strategic assets for the business and present opportunities for loyalty programmes, exclusive offers, and personalized marketing campaigns designed to increase long-term customer lifetime value.

Operational analysis using order status information provided insight into fulfilment performance. Monitoring order completion rates enables management to identify operational bottlenecks, improve delivery processes, and enhance overall customer satisfaction.

BUSINESS VALUE OF THE DASHBOARD

The completed dashboard transforms raw transactional records into meaningful business intelligence by presenting complex information through simple and interactive visualizations. Rather than manually reviewing hundreds of transactions, management can instantly monitor organizational performance using KPI cards, charts, and slicers that update dynamically based on selected filters.

The executive dashboard supports rapid decision-making by providing immediate visibility into revenue performance, customer trends, marketing effectiveness, and operational efficiency. Meanwhile, the customer and marketing dashboard offers deeper analytical

capabilities for evaluating promotional campaigns, referral channels, and customer purchasing behaviour.

The dashboard is designed to be scalable and can easily accommodate additional transaction data without requiring structural modifications, making it suitable for ongoing operational reporting.

RECOMMENDATIONS

Based on the analysis conducted, several strategic recommendations are proposed.

The organization should continue investing in high-performing marketing channels, particularly Instagram, while reassessing underperforming referral programmes to improve customer acquisition. Promotional campaigns should be reviewed regularly to identify the most effective coupon codes, allowing future marketing efforts to focus on promotions that generate the highest return on investment.

Given the relatively high number of unique customers compared to repeat purchases, customer retention initiatives such as loyalty programmes, personalized recommendations, and targeted email marketing should be implemented to encourage repeat business. Increasing customer lifetime value is likely to produce more sustainable revenue growth than relying solely on new customer acquisition.

The business should also leverage its high-value customer segment by offering exclusive incentives, VIP rewards, and early access to new products. Maintaining strong relationships with these customers can significantly improve long-term profitability.

Operationally, management should continuously monitor order fulfilment performance to reduce delays and improve delivery efficiency. Improving operational performance will enhance customer satisfaction, strengthen brand reputation, and increase repeat purchases.

Finally, dashboard reporting should become part of routine business reviews, enabling management to monitor key performance indicators regularly and make timely, evidence-based strategic decisions.

CONCLUSION

This project demonstrates how Microsoft Excel can be effectively utilized as a business intelligence platform for analysing transactional data and supporting strategic decision-making. Through structured data preparation, meaningful KPI development, interactive dashboard

design, and insightful visualization, the project successfully transformed raw operational data into valuable business information.

The completed dashboard provides management with a comprehensive overview of sales performance, customer behaviour, marketing effectiveness, and operational efficiency. More importantly, it enables faster decision-making by presenting complex analytical information in a clear, interactive, and user-friendly format. By implementing the recommendations identified through this analysis, the organization will be well positioned to strengthen customer relationships, optimize marketing investments, improve operational performance, and sustain long-term business growth.